



Optimizing Lipid-Based Delivery of Short Interfering RNA for Euchromatic Histone Lysine Methyltransferase 2 Gene Silencing in Mammalian Cells

Joshua Wu, Valerie Wu, Matthew To, Angela Wang
Lab Launch, Monrovia CA, USA

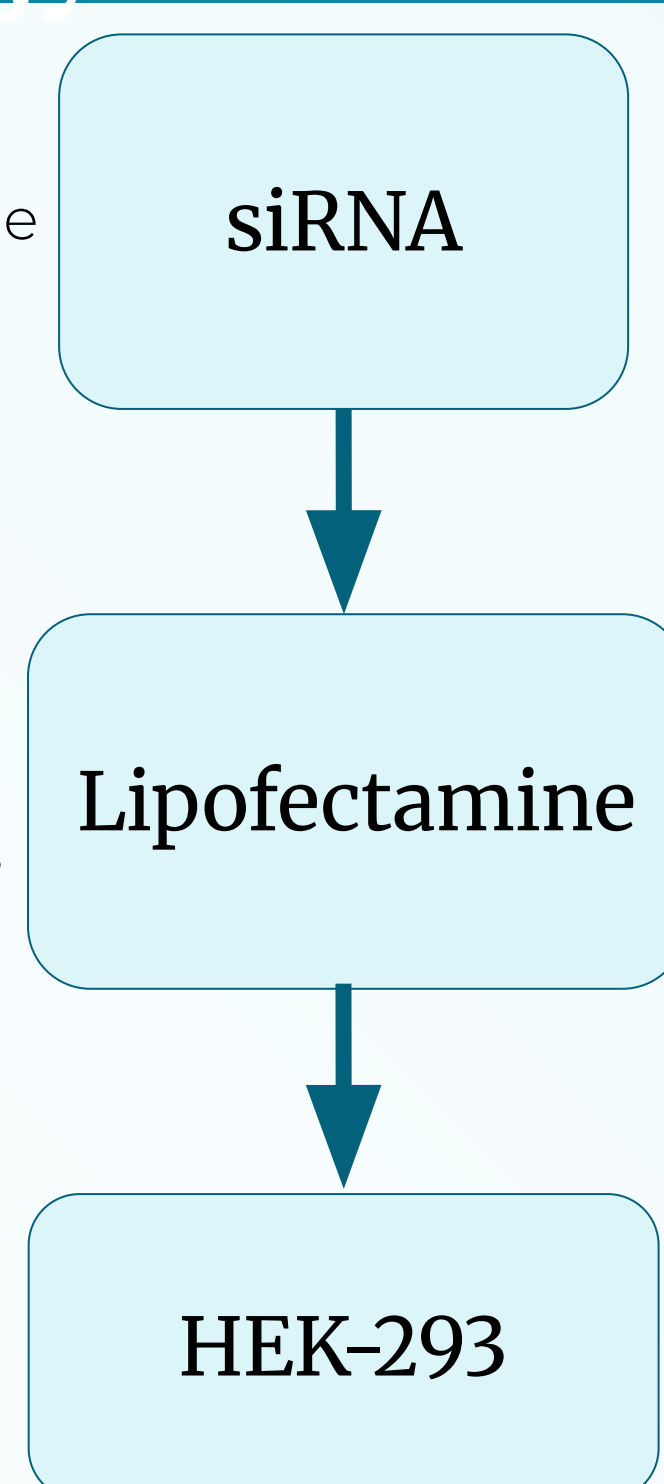


Introduction

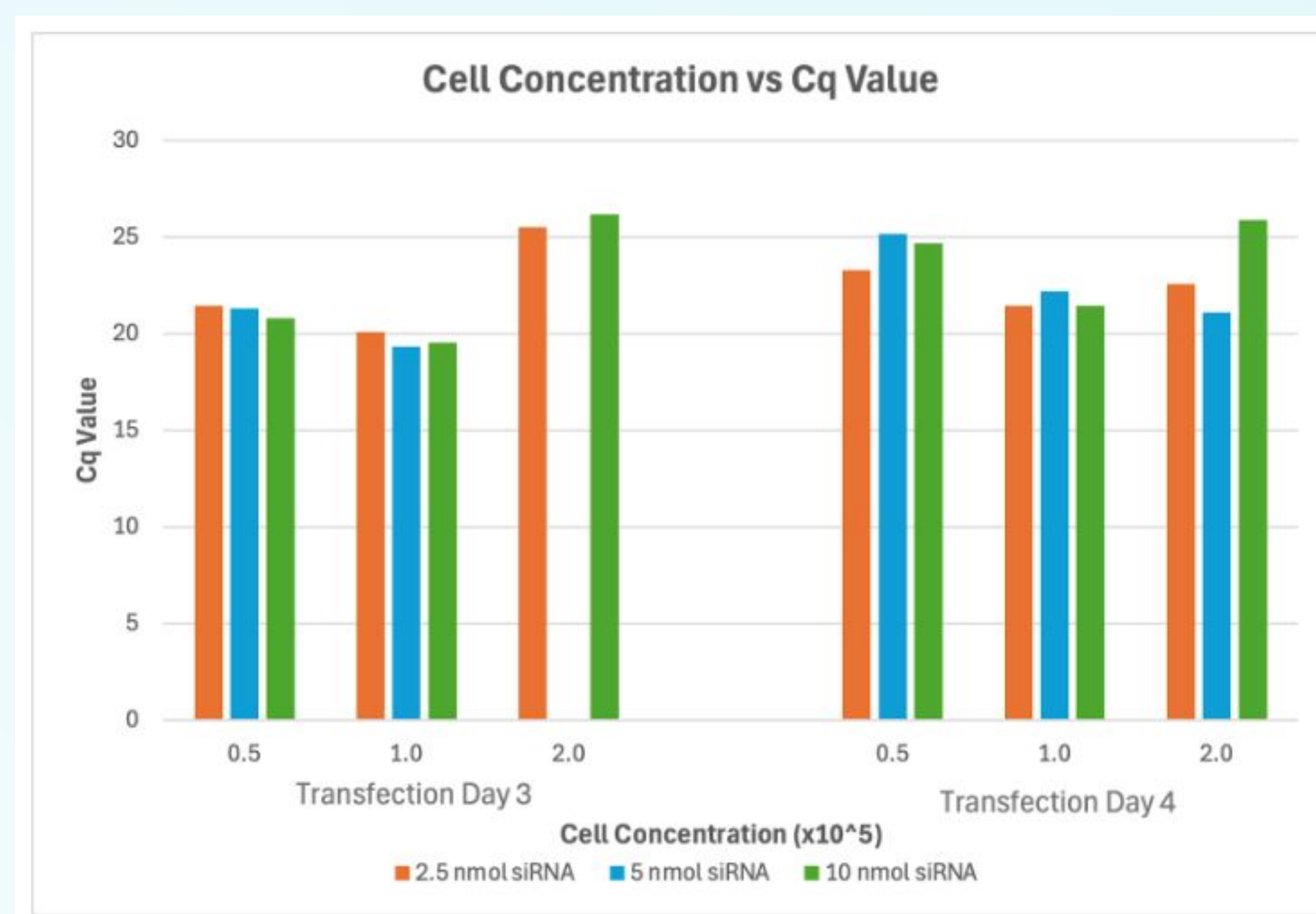
Substance use disorders represent a rapidly escalating global health crisis, with several studies continuously reporting alarming increases across diverse populations. In Los Angeles County alone, 7.6% of residents aged 12 and older suffer from chronic substance addiction, reflecting a broader national trend that contributes to rising rates of hospitalization and premature mortality. Despite the prevalence of this issue, existing interventions like behavioral therapies and pharmacological agents often have limited efficacy and high relapse rates. New molecular treatments of substance abuse focus on targeting specific proteins involved with addiction and silencing its mRNA expression before translation through short interfering RNA (siRNA) therapeutics. The epigenetic enzyme, G9a (encoded by EHMT2 mRNA), is responsible for the regulation of various dopamine-releasing brain circuits associated with substance consumption. This regulation occurs through G9a's methylation of histone H3 at lysine 9 within the adult nucleus accumbens (NAc), stimulating a positive-feedback loop and triggering the urge for greater substance consumption when G9a is in excess.

Methodology

Our team used Gibson Assembly to design a bacterial plasmid with the insertion of the EHMT2 gene, mimicking addiction through the stimulated overexpression of G9a. Gel electrophoresis was run to confirm successful cuts of the plasmid were made. This plasmid was then chemically transfected into the mammalian cell line HEK-293 by pairing plasmids with Lipofectamine 2000. Following the central dogma of gene expression, our team hypothesized that siRNA could be used to lower G9a concentrations by silencing G9a's EHMT2 mRNA. In these transformed cells, we tested for optimal concentrations of EHMT2 siRNA when paired with Lipofectamine RNAiMAX (a transfection reagent used to deliver siRNA), EHMT2 siRNA's efficacy in silencing mRNA, and the effect of varying cell concentrations in relation to the amount of EHMT2 mRNA silenced.

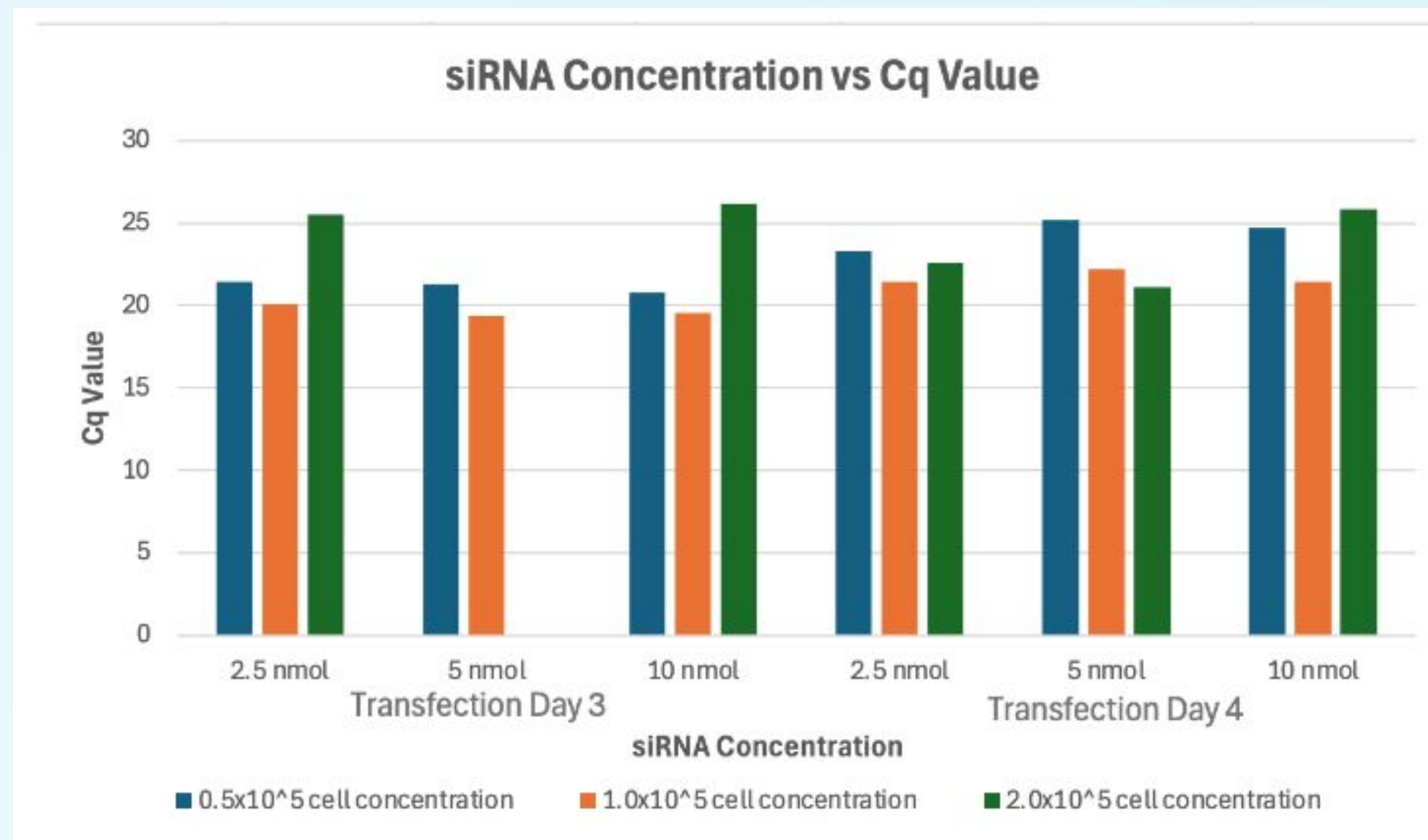


Result



HEK-293 cell concentration per well (0.5×10^5 , 1.0×10^5 , and 2.0×10^5 cells in a 96-well plate) versus qPCR Cq values on Transfection Days 3 and 4. Cells were treated with 2.5, 5, or 10 nmol siRNA. Lower Cq values indicate stronger target knockdown, while higher Cq values suggest reduced siRNA efficiency. Variations across siRNA dose and cell density highlight the influence of transfection conditions on gene silencing.

We reported that siRNA can successfully be utilized to decrease EHMT2 mRNA concentrations when in excess, preventing the onset of G9a overproliferation and dopamine-stimulated alcohol consumption. In our in vitro assay of siRNA paired with Lipofectamine encapsulation, transfection into HEK-293 mammalian cells resulted in decreased levels of EHMT2 mRNA production throughout the first week of observations. Compared to positive control blank wells where no siRNA was added, wells with siRNA added consistently had more gene silencing. Peak efficacy was then observed on days three and four, indicated with higher quantitative Polymerase Chain Reaction (qPCR) quantification cycle (Cq) values. We noted that increasing concentrations of siRNA greater than 2.5 nmol did not significantly impact the silencing of EHMT2 mRNA, suggesting a saturation point of Lipofectamine for EHMT2's siRNA to be lower than the typical 3.0-5.0 nmol amount of siRNA used in 300 μ L of Lipofectamine. Moreover, there was no effect on G9a silencing when comparing the cell concentrations of 0.5×10^5 , 1.0×10^5 , 2.0×10^5 cells per well, disproving our previous notion that EHMT2's siRNA efficacy is dependent on the amount of EHMT2 mRNA present. Using these results, we believe that for future clinical use, a siRNA therapeutic administered through a lipid nanoparticle delivery system would shield the siRNA from degradation in the body for effective silencing of EHMT2. This would establish a viable therapeutic option for treating substance abuse by silencing EHMT2 mRNA and down-regulating excessive G9a expression. Further confirmation of G9a's protein concentration is required through the means of western blotting, and future studies should consider modifying the lipid nanoparticle delivery system to successfully reach the demethylation site of NAc.



siRNA concentration (2.5, 5, and 10 nmol) versus qPCR Cq values in HEK-293 cells transfected at different cell densities (0.5×10^5 , 1.0×10^5 , and 2.0×10^5 cells per well) on Transfection Days 3 and 4. Lower Cq values indicate stronger gene silencing, while higher Cq values suggest reduced siRNA efficacy. The data illustrate how both siRNA dosage and cell concentration influence transfection efficiency across time points.

Future Directions

- Transcriptome-wide off-target profiling via RNA-seq to detect unintended gene silencing.
- Assess innate immune activation (TLR3/7, cytokines) following siRNA-Lipofectamine exposure.
- Compare chemical siRNA modifications (2'-OMe, 2'-F, phosphorothioates) for stability and potency.
- Add endosomal-escape enhancers; quantify cytosolic delivery versus lysosomal sequestration.
- Test patient-derived iPSC dopaminergic neurons to verify target engagement and translatability.
- Map H3K9me2 landscape changes by ChIP-seq; evaluate durability and reversibility post-washout.
- Benchmark Lipofectamine against clinical-grade LNPs and exosomes for CNS delivery.
- Engineer ligand-targeted carriers (transferrin/apoE peptides) to cross BBB and home to NAc.
- Build PK/PD models: siRNA half-life, knockdown kinetics, and optimal redosing intervals.
- Run comprehensive tox panels—hepatic, renal, neurobehavioral—under repeated-dose regimens.

References

- NIDA. "Most reported substance use among adolescents held steady in 2022." National Institute on Drug Abuse, 15 Dec. 2022. <https://nida.nih.gov/news-events/news-releases/2022/12/most-reported-substance-use-among-adolescents-held-steady-in-2022>
- NIDA. "Treatment and Recovery." National Institute on Drug Abuse, 25 Sep. 2023. <https://nida.nih.gov/publications/drugs-brains-behavior-science-addiction/treatment-recovery>
- NIDA. "Drugs and the Brain." National Institute on Drug Abuse, 1 Jul. 2011. <https://nida.nih.gov/publications/drugs-brains-behavior-science-addiction/drugs-brain>
- Nielsen, David A et al. "Epigenetics of drug abuse: predisposition or response." *Pharmacogenomics* vol. 13,10 (2012): 1149-60. doi:10.2217/pgs.12.94.
- LNP Main sources: Hagino, Yuta et al. "GALA-Modified Lipid Nanoparticles for the Targeted Delivery of Plasmid DNA to the Lungs." *Molecular pharmaceutics* vol. 18,3 (2021): 878-888. doi:10.1021/acs.molpharmaceut.0c00854
- Pardridge, William M. "Brain Gene Therapy with Trojan Horse Lipid Nanoparticles." *Trends in Molecular Medicine*, vol. 29, no. 5, 2023, pp. 343-353.
- Tuma, Jan et al. "Lipid Nanoparticles Deliver mRNA to the Brain after an Intracerebral Injection." *Biochemistry* vol. 62,24 (2023): 3533-3547. doi:10.1021/acs.biochem.3c00371
- Cheng, Qiang et al. "Selective organ targeting (SORT) nanoparticles for tissue-specific mRNA delivery and CRISPR-Cas gene editing." *Nature nanotechnology* vol. 15,4 (2020): 313-320. doi:10.1038/s41565-020-0669-6
- Chatterjee, Sushmita et al. "Endosomal escape: A bottleneck for LNP-mediated therapeutics." *Proceedings of the National Academy of Sciences of the United States of America* vol. 121,11 (2024): e2307800120. doi:10.1073/pnas.2307800120
- Chu, Runxuan et al. "Lipid Nanoparticles as the Drug Carrier for Targeted Therapy of Hepatic Disorders." *National Advanced Medical Engineering Research Center, China State Institute of Pharmaceutical Industry*, 24 Apr. 2024.
- Macki, Masatoshi et al. "Production of siRNA-Loaded Lipid Nanoparticles using a Microfluidic Device." *Journal of visualized experiments : JoVE*, 181 10.3791/62999. 22 Mar. 2022. doi:10.3791/62999.13060

Acknowledgments

Thermo Fisher Scientific
HEK-293 cells supplied by ATCC
Lab Launch Beckman Research Institute of City of Hope